

Mingxing (Ethan) Rao

Curriculum Vitae

✉ mingxing.rao@vanderbilt.edu
🌐 [mx-ethan-rao.github.io](https://github.com/mx-ethan-rao)
in [mingxing-rao-488030261](https://www.linkedin.com/in/mingxing-rao-488030261)
🔗 [mx-ethan-rao](https://www.researchgate.net/profile/Mingxing-Rao)
🔗 [nAJns40AAAAJ&hl](https://www.kaggle.com/nAJns40AAAAJ&hl)

Education

- Jan. 2023 – Nov 2027 **Ph.D. in Computer Science**, *Vanderbilt University*, Nashville, USA
(Anticipated) Advisor: Daniel Moyer; GPA: 3.76/4.0
- Sep. 2018 – May 2022 **B.Sc. (Hons.) in Computer Science**, *Hong Kong Baptist University*, Hong Kong, China
Scholastic Award(Top 2% academic performance of graduates); GPA: 3.7/4.0

Research Interests

Generative Models, Model Inversion, Membership Inference, Monocular Depth Estimation, Representation Learning

Selected Publications

- [ECCV 2026] M. Rao, D. Moyer. *Generalization and Memorization in Rectified Flow*. The European Conference on Computer Vision (2026). Preprint: <https://arxiv.org/pdf/2603.13421v1>
- [CVPR 2026] M. Rao, B. Qu, D. Moyer. *Latent Diffusion Inversion Requires Understanding the Latent Space*. The IEEE/CVF Conference on Computer Vision and Pattern Recognition (2026). Preprint: <https://arxiv.org/pdf/2511.20592>
- [TMLR] M. Rao, B. Qu, D. Moyer. *Score-based Membership Inference on Diffusion Models*. Transactions on Machine Learning Research. Preprint: <https://arxiv.org/pdf/2509.25003>
- M. Rao, B. Jiang, D. Moyer. *Training Noise Token Pruning*. Preprint: <https://arxiv.org/pdf/2411.18092>
- [IPCAI 2024 & IJCARS] M. Rao, Y. Qin, S. Kolouri, J.Y. Wu, D. Moyer. *Zero-shot Prompt-based Video Encoder for Surgical Gesture Recognition*. International Journal of Computer Assisted Radiology and Surgery (2024). Paper: <https://link.springer.com/article/10.1007/s11548-024-03257-1>
- [Poster at RECOMB 2023] Z. Zhang[†], M. Rao[†], Y. Huang, L. Zhang. *A graph-based Gaussian Mixture Variational Autoencoder improves metagenome binning for short contigs*. International Conference on Research in Computational Molecular Biology (2023). DOI: <https://doi.org/10.21203/rs.3.rs-3308172/v1>

Experience

- Jan. 2026 – Present **Research Assistant**, *The Advanced Research Projects Agency for Health (ARPA-H)*, Nashville, US
- Work on the robotic vision development for an automated surgical program, specializing in Monocular Depth Estimation (MDE) for endoscopic imaging.
 - Manage the end-to-end data pipeline, from collection to model architecture design, focusing on high-accuracy absolute depth perception during Central Airway Obstruction (CAO) and Benign Prostatic Hyperplasia (BPH) procedures.
- Jan. 2023 – Present **Research Assistant**, *Vanderbilt University*, Nashville, US
- Investigate efficient machine learning, specifically token pruning for Vision Transformers (ViT).
 - Investigate the robustness of generative models against potential vulnerabilities..
- Apr. 2022 – Nov. 2022 **Machine Learning Engineer Intern**, *Baidu NLP Group*, Shenzhen, China
- Developed protein-ligand binding simulation pipelines with Gromacs for data augmentation.
 - Designed GNN model improving affinity prediction accuracy on PDBBind.

- May 2021 – Jul. 2021 **Research Intern**, *National University of Singapore*, Singapore
- Built academic knowledge graph: entity extraction, graph-based fusion, rule-based inference.
 - Conducted blockchain data mining: built graphs from chain data.
- Sep. 2021 – Dec. 2022 **Undergraduate Researcher**, *Hong Kong Baptist University*, Hong Kong, China
- Improved metagenomic contig binning using Gaussian Mixture VAEs with graph embeddings.
 - Results published as co-first author in poster to **RECOMB 2023**.

Teaching

- 2025 Spring **Teaching Assistant**, *CS4262 Foundations of Machine Learning*, Vanderbilt University
- 2024 Fall **Teaching Assistant**, *CS4260 Artificial Intelligence*, Vanderbilt University
- 2024 Spring **Teaching Assistant**, *CS8395 Representation Learning*, Vanderbilt University
- 2023 Fall **Teaching Assistant**, *DS5460 Big Data Scaling*, Vanderbilt University

Awards

- 2022 Scholastic Award (Top 2% academic performance of graduates)
- 2021 C.V. Starr Scholarship Fund (~35,000 HKD)
- 2021 Faculty of Science Overseas Research Scholarship for Outstanding Students in 2020-2021 (~30,000 HKD)
- 2019 Mr. & Mrs. Lau Chor Tak Scholarship Scheme for Outstanding Students (~75,000 HKD)

Skills

Programming: Python, PyTorch, TensorFlow, JAX, Java, C/C++, CUDA, Git, Linux, Docker

Languages: English, Chinese